

## **What to Do About Side Effects**

- Wait
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- In a few weeks, switch to another agent or add other drugs

## **Best Augmenting Agents for Side Effects**

- Often best to try another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for insomnia
- Bupropion, sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Bupropion for emotional flattening, cognitive slowing, or apathy
- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Benzodiazepines for jitteriness and anxiety, especially at initiation of treatment and especially for anxious patients
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium,

a mood stabilizer or an atypical antipsychotic, and/or  
discontinuation of vilazodone

## **Dosing And Use**

### **Usual Dosage Range**

- 20–40 mg/day

### **Dosage Forms**

- Tablet 10 mg, 20 mg, 40 mg

### **How To Dose**

- Initial 10 mg/day; increase to 20 mg/day after one week; can increase to 40 mg/day after one more week; should be taken with food



### **Dosing Tips**

- Given once daily, any time of day tolerated but must be administered with food, because taking on an empty stomach may reduce absorption of vilazodone by 50%
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activating a bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

- No minimally effective dose has been established, so theoretically possible that doses lower than 40 mg daily may be effective in some patients
- Doses higher than 40 mg daily have not been well studied, but theoretically possible that slow titration to doses higher than 40 mg may be effective in some patients, particularly those with treatment-resistant depression who fail to respond adequately to 40 mg daily

### **Overdose**

- Few reports of vilazodone overdose
- No fatalities; serotonin syndrome, lethargy, restlessness, hallucinations, disorientation

### **Long-Term Use**

- Has not been evaluated in controlled studies, but long-term treatment of major depressive disorder is generally necessary

### **Habit Forming**

- No

### **How to Stop**

- Tapering to avoid potential withdrawal reactions generally prudent
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation

- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

### **Pharmacokinetics**

- Mean terminal half-life 25 hours
- Metabolized by CYP450 3A4
- Absorption and bioavailability are reduced by half when taken on an empty stomach

### **Drug Interactions**

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOI or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing vilazodone
- Inhibitors of CYP450 3A4, such as nefazodone, fluoxetine, fluvoxamine, and even grapefruit juice, may decrease the clearance of vilazodone and thereby raise its plasma levels, so dose should be reduced to 20 mg when coadministered with strong CYP3A4 inhibitors
- Inducers of CYP450 3A4, such as carbamazepine, may increase clearance of vilazodone and thus lower its plasma levels and

possibly reduce therapeutic effects

- Could theoretically cause weakness, hyperreflexia, and incoordination when combined with sumatriptan or possibly other triptans, requiring careful monitoring of patient
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)

### **Other Warnings/Precautions**

- Use with caution in patients with history of seizure
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- Not approved in children, so when treating children off label, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

### **Do Not Use**

- If patient is taking an MAOI
- If there is a proven allergy to vilazodone

## **Special Populations**

### **Renal Impairment**

- No dose adjustment necessary

### **Hepatic Impairment**

- No dose adjustment necessary for mild to moderate impairment
- Has not been studied in patients with severe hepatic impairment

### **Cardiac Impairment**

- Not systematically evaluated in patients with cardiac impairment
- Vilazodone has not shown any significant effect on blood pressure, heart rate, or QT interval in placebo-controlled trials
- Treating depression with SSRIs in patients with acute angina or following myocardial infarction may reduce cardiac events and improve survival as well as mood

### **Elderly**

- No dose adjustment necessary
- Some patients may tolerate lower doses better

- Risk of SIADH with SSRIs is higher in the elderly
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older

### **Children and Adolescents**

- Safety and efficacy have not been established
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and strongly consider informing parents or guardians of this risk so they can help observe child or adolescent patients

### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Nonetheless, continuous treatment during pregnancy may be necessary and has not been proven to be harmful to the fetus

- At delivery there may be more bleeding in the mother and transient irritability or sedation in the newborn
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients, this may mean continuing treatment during pregnancy
- Exposure to serotonin reuptake inhibitors early in pregnancy may be associated with increased risk of septal heart defects (absolute risk is small)
- Serotonin reuptake inhibitor use beyond the 20th week of pregnancy may be associated with increased risk of pulmonary hypertension in newborns, although this is not proven
- Exposure to serotonin reuptake inhibitors late in pregnancy may be associated with increased risk of gestational hypertension and preeclampsia
- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying

- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

### **Breast Feeding**

- Unknown if vilazodone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Trace amounts may be present in nursing children whose mothers are on vilazodone
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

## **The Art Of Psychopharmacology**

### **Potential Advantages**

- Patients with sexual dysfunction on an SSRI/SNRI or who wish to avoid sexual dysfunction on an antidepressant
- Patients with weight gain on another antidepressant or who wish to avoid weight gain on an antidepressant
- Patients with mixed anxiety and depression

### **Potential Disadvantages**

- Patients who cannot take medication reliably with food
- Patients sensitive to gastrointestinal side effects such as diarrhea and nausea

### **Primary Target Symptoms**

- Depressed mood
- Anxiety

### **Pearls**

- Serotonin 1A partial agonist property is a relatively unique mechanism of action among approved antidepressants (also vortioxetine and atypical antipsychotic augmenting agents such as quetiapine, aripiprazole, brexpiprazole, and others)
- First member of a new antidepressant class, SPARIs, or serotonin partial agonist reuptake inhibitors
- Relative lack of sexual dysfunction and weight gain compared to many other antidepressants that block serotonin reuptake may be

due to the serotonin 1A partial agonist properties of vilazodone

- High doses would theoretically raise brain serotonin levels more robustly than the standard dose, and may improve efficacy in some patients but reduce tolerability in some patients
- Consider doses of 50–80 mg daily if effective and well tolerated for patients with treatment-resistant depression or treatment-resistant OCD and other anxiety disorders
- Consider for patients with comorbid anxiety disorders plus depression
- Nonresponse to vilazodone in elderly may require consideration of mild cognitive impairment or Alzheimer disease

### *Suggested Reading*

Citrome L. Vilazodone for major depressive disorder: a systematic review of the efficacy and safety profile for this newly approved antidepressant – what is the number needed to treat, number needed to harm and likelihood to be helped or harmed? *Int J Clin Pract* 2012 ;**66** (4):356–68 .

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Dawson LA , Watson JM. Vilazodone: a 5HT1A receptor agonist/serotonin transporter inhibitor for the treatment of affective disorders . *CNS Neurosci Therapeutics* 2009 ;**15** :107–17 .

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Laughren TP , Gobburu J , Temple RJ , et al. Vilazodone: clinical basis for the US Food and Drug Administration’s approval of a new antidepressant .

*J Clin Psychiatry* 2011 ;**72** (9):1166 –73.

.....

Rickels K , Athanasiou M , Robinson D , et al. Evidence for efficacy and tolerability of vilazodone in the treatment of major depressive disorder: a randomized, double-blind, placebo-controlled trial . *J Clin Psychiatry* 2009 ;e1 –e8.

Vortioxetine

## Therapeutics

### Brands

- Trintellix (formerly Brintellix)
- *see index for additional brand names*

### Generic?

No

### Class

- Neuroscience-based Nomenclature: serotonin multimodal (S-MM)
- Multimodal antidepressant

### Commonly Prescribed for

*(bold for FDA approved)*

- **Major depressive disorder**
- Generalized anxiety disorder (GAD)
- Cognitive symptoms associated with depression
- Geriatric depression

### How the Drug Works

- Increases release of several different neurotransmitters (serotonin, norepinephrine, dopamine, glutamate, acetylcholine, and histamine) and reduces the release of GABA through 3 different modes of action
- Mode 1: blocks serotonin reuptake pump (serotonin transporter)
- Mode 2: binds to G protein-linked receptors (full agonist at serotonin 1A receptors, partial agonist at serotonin 1B receptors, antagonist at serotonin 1D and serotonin 7 receptors)
- Mode 3: binds to ion channel-linked receptors (antagonist at serotonin 3 receptors)
- Full agonist actions at presynaptic somatodendritic serotonin 1A autoreceptors may theoretically enhance serotonergic activity and contribute to antidepressant actions
- Full agonist actions at postsynaptic serotonin 1A receptors may theoretically diminish sexual dysfunction caused by serotonin reuptake inhibition
- Antagonist actions at serotonin 3 receptors may theoretically enhance noradrenergic, acetylcholinergic, and glutamatergic activity and contribute to antidepressant and pro-cognitive actions
- Antagonist actions at serotonin 3 receptors may theoretically reduce nausea and vomiting caused by serotonin reuptake inhibition
- Antagonist actions at serotonin 7 receptors may theoretically contribute to antidepressant and pro-cognitive actions as well as reduce insomnia caused by serotonin reuptake inhibition

- Partial agonist actions at serotonin 1B receptors may enhance not only serotonin release, but also acetylcholine and histamine release
- Antagonist actions at serotonin 1D receptors may enhance serotonin release and may also theoretically enhance the release of pro-cognitive neurotransmitters and thereby enhance pro-cognitive actions

### **How Long Until It Works**

- Onset of therapeutic actions is usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6 or 8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

### **If It Works**

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression

- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

### **If It Doesn't Work**

- Many patients only have a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider switching to another agent or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and switch to a mood stabilizer



### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- Augmentation experience is limited compared to other antidepressants
- Trazodone, especially for insomnia
- Bupropion, mirtazapine, reboxetine, or atomoxetine (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Use with caution with antidepressants that are CYP450 2D6 inhibitors (e.g., bupropion, duloxetine, fluoxetine, paroxetine), as these agents will increase vortioxetine levels and may require a dose reduction of vortioxetine
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, treatment-resistant depression, or treatment-resistant anxiety disorders
- Benzodiazepines
- If all else fails for anxiety disorders, consider gabapentin, pregabalin, or tiagabine
- Hypnotics for insomnia
- Classically, lithium, buspirone, or thyroid hormone

## Tests

- None for healthy individuals

# Side Effects

## How Drug Causes Side Effects

- Theoretically due to increases in serotonin concentrations at serotonin receptors in parts of the brain and body other than those that cause therapeutic actions (e.g., unwanted actions of serotonin at central serotonin 1A receptors causing nausea, unwanted actions of serotonin in the CNS causing sexual dysfunction, etc.)
- Most side effects are immediate but often go away with time, in contrast to most therapeutic effects, which are delayed and are enhanced over time

## Notable Side Effects

- Nausea, vomiting, constipation
- Sexual dysfunction



## Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare induction of mania and activation of suicidal ideation

## Weight Gain



- Reported but not expected

## Sedation



- Reported but not expected

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- Trazodone or a hypnotic for insomnia
- Sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Bupropion for emotional flattening, cognitive slowing, apathy, or sexual dysfunction (with caution, as bupropion can raise vortioxetine levels via CYP450 2D6 inhibition)
- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Benzodiazepines for jitteriness and anxiety, especially at initiation of treatment and especially for anxious patients
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)

- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of vortioxetine

## **Dosing And Use**

### **Usual Dosage Range**

- 5–20 mg/day

### **Dosage Forms**

- Tablet 5 mg, 10 mg, 15 mg, 20 mg

### **How to Dose**

- Initial 10 mg once daily; can decrease to 5 mg once daily or increase to 20 mg once daily depending on patient response; maximum recommended dose generally 20 mg once daily



### **Dosing Tips**

- Can be taken with or without food
- Tablet should not be divided, crushed, or dissolved

- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activating a bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

### **Overdose**

- No fatalities have been reported; nausea, dizziness, diarrhea, abdominal discomfort, generalized pruritus, somnolence, flushing

### **Long-Term Use**

- Long-term treatment of major depressive disorder is generally necessary

### **Habit Forming**

- No

### **How to Stop**

- Taper not necessary with recommended doses

### **Pharmacokinetics**

- Metabolized by CYP450 2D6, 3A4/5, 2C19, 2C9, 2A6, 2C8, and 2B6
- Mean terminal half-life approximately 66 hours

## **Drug Interactions**

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14–21 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (about 14 days for vortioxetine with a half-life of 66 hours) after discontinuing vortioxetine
- Strong CYP450 2D6 inhibitors can increase plasma levels of vortioxetine, possibly requiring its dose to be decreased
- Broad CYP450 2D6 inducers can decrease plasma levels of vortioxetine, possibly requiring its dose to be increased
- Could theoretically cause weakness, hyperreflexia, and incoordination when combined with sumatriptan or possibly other triptans, requiring careful monitoring of patient
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)

## **Other Warnings/Precautions**

- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent

- Possible risk of hyponatremia related to SIADH (syndrome of inappropriate antidiuretic hormone secretion) with serotonergic drugs
- Not approved in children, so when treating children off label, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of non-treatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

### **Do Not Use**

- If patient is taking an MAOI
- If there is a proven allergy to vortioxetine

## **Special Populations**

### **Renal Impairment**

- No dose adjustment necessary

### **Hepatic Impairment**

- No dose adjustment necessary for mild to moderate impairment
- Has not been studied in patients with severe hepatic impairment

### **Cardiac Impairment**

- Not systematically evaluated in patients with cardiac impairment
- Treating depression with SSRIs in patients with acute angina or following myocardial infarction may reduce cardiac events and improve survival as well as mood; not known for vortioxetine

### **Elderly**

- No dose adjustment necessary
- Some patients may tolerate lower doses better
- Risk of SIADH with SSRIs is higher in the elderly
- Reduction in risk of suicidality with antidepressants compared to placebo in adults age 65 and older



### **Children and Adolescents**

- Safety and efficacy have not been established
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart

- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and strongly consider informing parents or guardian of this risk so they can help observe child or adolescent patients

### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Nonetheless, continuous treatment during pregnancy may be necessary and has not been proven to be harmful to the fetus
- At delivery there may be more bleeding in the mother and transient irritability or sedation in the newborn
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no